BioMaster

AI-assisted drug repurposing workflow for cancer-oriented candidate prioritization

External project report May 2026

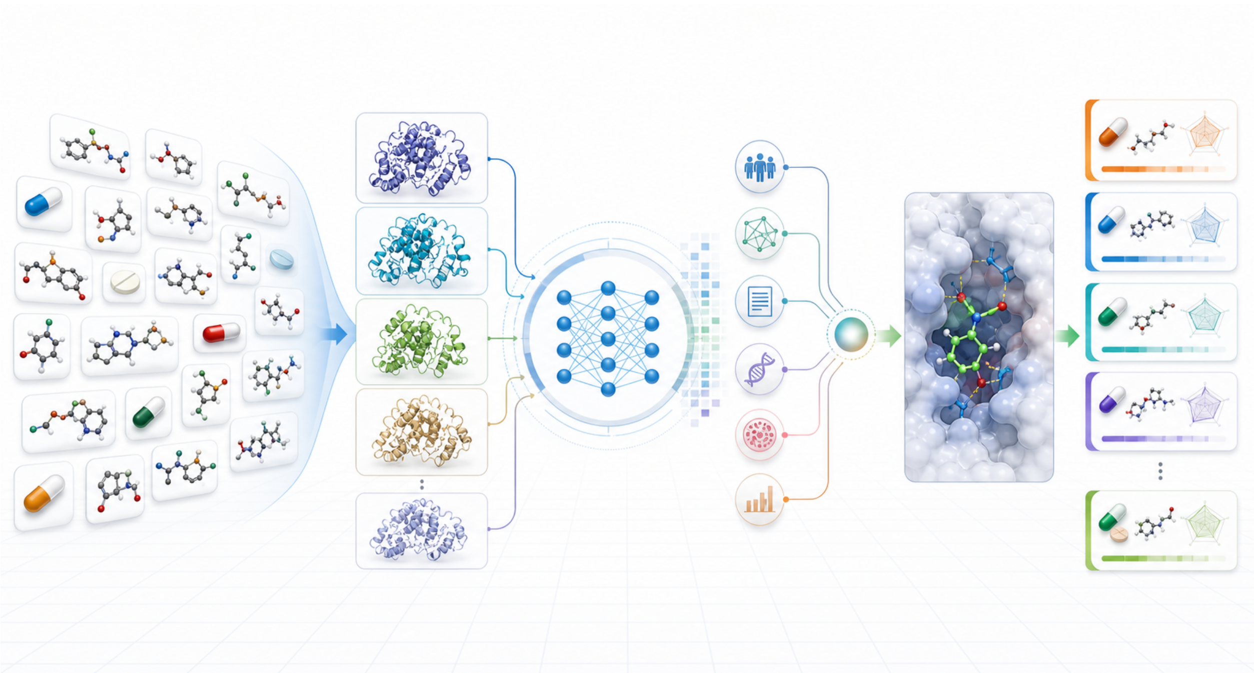

Figure 1. BioMaster computational screening workflow. The main figure was generated with GPT-image-2 to illustrate the integrated workflow from approved small-molecule drugs and human protein targets to AI-DTI prediction, disease-evidence fusion, structural docking enhancement, and ranked candidate output. Quantitative results and methodological details are provided in the report body.

This report summarizes the computational analysis, screening rationale, evidence-integration strategy, workflow design, and candidate-prioritization output for downstream scientific review and validation planning.

1. Executive Summary

BioMaster was developed as a reproducible computational workflow for drug repurposing. The current analysis focuses on a cancer-oriented discovery task: identifying approved small-molecule drugs and human protein targets that warrant further mechanistic review and experimental validation.

The workflow evaluates a broad drug-target search space and then progressively adds biological context. First, approved drugs and human proteins are paired at scale for AI-based drug-target interaction prediction. Second, cancer-relevance evidence is incorporated at the target and protein-network levels. Third, drug-disease graph evidence is added where reliable mappings are available. Finally, the highest-priority candidates are structurally evaluated to support interpretation of possible binding modes.

915	1000	915000	1000
approved small molecules	human protein targets	screened drug-target pairs	structure-prioritized pairs

The primary output is a ranked set of computationally prioritized drug-target pairs. These results are intended to support candidate selection, literature review, mechanism generation, and experimental planning. They should be interpreted as prioritization hypotheses rather than experimental proof of therapeutic efficacy.

2. Study Objective and Scope

The project was designed to answer three practical questions:

1. Which approved small-molecule drugs show predicted interaction potential with human protein targets?
2. Which predicted target proteins carry cancer-relevant biological evidence from curated disease databases and protein interaction networks?
3. Which high-priority drug-target pairs also show structure-level support that can guide mechanistic interpretation and validation design?

The disease scope in this report is pan-cancer. The workflow uses cancer as a broad disease concept rather than a single tumor type, molecular subtype, or mutation-defined patient population. This design is appropriate for establishing a general candidate-discovery framework. A narrower disease scope can be incorporated later by replacing the disease concept used in the evidence-fusion layer.

3. Work Completed

3.1 Approved Drug Library Construction

A library of 915 approved small-molecule drugs was curated and standardized. Drug records were harmonized across ChEMBL and PubChem where available. Each compound was prepared with model-ready molecular information, including standardized identifiers, SMILES, InChIKey, molecular formula, and structural files for downstream docking preparation.

3.2 Human Protein Target Library Construction

A target library of 1000 human proteins was constructed. Each target was represented by UniProt accession, gene symbol, protein name, amino-acid sequence, and available AlphaFold structural information. This target library provided the protein-side input for sequence-based drug-target prediction and structure-based docking analysis.

3.3 Large-Scale AI-DTI Screening

The drug and target libraries were crossed to generate 915000 drug-target pairs. ConPLex was used as the scalable AI-DTI prediction layer to estimate interaction potential from drug molecular representation and protein sequence. This step provided the broad interaction signal and avoided prematurely narrowing the analysis to a small manually selected target set.

3.4 Cancer-Relevance Evidence Integration

The AI-DTI results were combined with target-level cancer evidence from Open Targets and network-level evidence from STRING. Open Targets contributes curated target-disease association scores, while STRING contributes protein interaction context that helps capture disease-relevant network proximity.

This evidence is target-level evidence. It means that a gene or protein is associated with cancer in curated databases or disease-related interaction networks. It does not imply that every screened protein contains a cancer-specific mutation site. It also does not imply that an approved drug has already demonstrated clinical efficacy through that target.

3.5 Drug-Disease Graph Evidence Integration

TxGNN was used to add a drug-disease graph signal. For drugs that could be mapped into the TxGNN score table, the drug-level cancer indication signal was incorporated into the final disease-oriented priority score. For unmapped drugs, the workflow retained the Open Targets and STRING target-evidence score; missing TxGNN mapping was not treated as negative evidence.

3.6 Structural Docking Enhancement

The highest-priority candidates were evaluated with DiffDock to obtain ligand-protein pose predictions and confidence information. This structural layer was used as an interpretive enhancement for selected candidates. It helps identify drug-target pairs that are supported not only by AI-DTI prediction and disease relevance, but also by plausible structure-level evidence for downstream mechanistic discussion.

4. Analytical Rationale

The central design principle is that each evidence source answers a different scientific question:

Evidence source	Scientific role in the workflow
ConPLex	Estimates whether a drug may interact with a protein using molecular and sequence information.
Open Targets	Provides curated target-disease association evidence for the cancer disease concept.
STRING	Provides protein interaction context to capture network-level disease relevance.
TxGNN	Adds drug-disease graph evidence when drug mappings are available.
DiffDock	Provides structure-level docking support for selected high-priority drug-target pairs.

The workflow therefore does not rely on a single model score. Candidate priority increases when independent evidence layers converge: predicted drug-target interaction, target-level cancer relevance, network support, drug-disease graph signal where available, and structural docking support.

5. Evidence-Fusion Strategy

5.1 Target-Evidence Priority

For each drug-target pair, the Open Targets and STRING priority score was computed as:

$$\begin{aligned} P_{\mathrm{OT+STRING}} &= 0.55 \times S_{\mathrm{AI}} \\ &+ 0.30 \times S_{\mathrm{OpenTargets}} \\ &+ 0.15 \times S_{\mathrm{STRING}} \end{aligned}$$

This formula places the largest weight on predicted drug-target interaction while reserving substantial weight for direct and network-based disease relevance.

5.2 Disease-Augmented Priority

When TxGNN mapping was available, the disease-augmented priority score was computed as:

$$\begin{aligned} P_{\mathrm{Disease}} &= 0.80 \times P_{\mathrm{OT+STRING}} \\ &+ 0.20 \times S_{\mathrm{TxGNN}} \end{aligned}$$

This step adds drug-level disease information while preserving target-specific evidence as the main ranking component.

5.3 Structure-Enhanced Consensus

For the structurally evaluated candidate set, DiffDock confidence was normalized and combined with the disease-augmented priority score:

$$\begin{aligned} P_{\mathrm{Consensus}} &= 0.85 \times P_{\mathrm{Disease}} \\ &+ 0.15 \times S_{\mathrm{DiffDock,norm}} \end{aligned}$$

This design treats docking as supportive structural evidence rather than as a replacement for disease relevance or predicted drug-target interaction.

6. Computational Workflow

Workflow layer	Role in the study
Drug standardization	Prepare approved small molecules with harmonized identifiers and molecular representations.
Protein target construction	Prepare human protein targets with sequence and structural context.
AI-DTI screening	Evaluate all approved drug and human protein combinations with ConPLex.
Disease-evidence fusion	Combine target-level cancer evidence from Open Targets and STRING with AI-DTI scores.
Drug-disease augmentation	Add TxGNN drug-disease graph evidence when available.
Structural enhancement	Use DiffDock on high-priority candidates to support pose-level interpretation.
Candidate prioritization	Produce ranked drug-target pairs for expert review and experimental planning.

7. Results Summary

Output	Result
Drug library	915 approved small molecules were standardized for model and structure workflows.
Target library	1000 human protein targets were prepared with sequence-level information.
AI-DTI prediction	915000 drug-target pairs were evaluated with ConPLex.
Disease-oriented ranking	915000 pairs were ranked with integrated AI-DTI, Open Targets, STRING, and TxGNN evidence.
Structure-enhanced candidate set	The prioritized structural candidate set contains 1000 drug-target pairs, covering 337 unique drugs and 64 unique proteins.
Focused review pool	The top-ranked 100 pairs cover 67 unique drugs and 20 unique proteins, creating a practical pool for expert review and experimental shortlist development.

The top-ranked candidates are enriched for targets with strong pan-cancer evidence, including EGFR, KIT, and JAK1. This pattern is consistent with the workflow design: a high-priority candidate must show predicted drug-target interaction potential and cancer-relevance evidence before receiving a strong final rank.

8. Representative Prioritized Candidates

The table below lists representative high-ranking pairs from the structure-enhanced consensus ranking.

Rank	Drug	Target	Disease score	Consensus score
1	Afatinib Dimaleate	EGFR	0.926858	0.917979
2	Dacomitinib	EGFR	0.889966	0.874261
3	Cabozantinib S-Malate	KIT	0.864188	0.864188
4	Momelotinib Dihydrochloride	EGFR	0.861371	0.856885
5	Repotrectinib	JAK1	0.860247	0.850873
6	Pazopanib Hydrochloride	KIT	0.901546	0.848898
7	Osimertinib Mesylate	EGFR	0.867670	0.839205
8	Trilaciclib Dihydrochloride	JAK1	0.797277	0.827686
9	Neratinib Maleate	EGFR	0.841782	0.808585
10	Tucatinib	EGFR	0.845578	0.806943

The candidate list should be read as a ranked computational hypothesis set. The strongest candidates are those where multiple independent evidence layers agree: predicted drug-target interaction, target-level cancer association, drug-disease graph signal where available, and structural docking support.

9. Interpretation of Cancer Evidence

Open Targets does not make every protein a cancer target. It provides target-disease association evidence for proteins that have curated genetic, literature, pathway, expression, somatic mutation, or related disease signals. Therefore, Open Targets contributes evidence only where such associations exist and are scored.

The cancer evidence used in this workflow is not mutation-site-specific evidence. A cancer-specific site, mutation hotspot, or altered binding pocket is usually relevant only for particular proteins, tumor types, and molecular contexts. In contrast, the current analysis uses target-level cancer association as a broad prioritization signal. This distinction is essential: a protein can be cancer-associated without every residue or binding site being cancer-specific.

The workflow is therefore best interpreted as a prioritization engine. It narrows a large computational search space into a smaller set of drug-target hypotheses that can be reviewed using tumor subtype knowledge, known mutations, pathway biology, drug safety, availability, novelty, and experimental feasibility.

10. Downstream Validation Framework

The next phase should convert ranked computational hypotheses into experimentally actionable candidates. The validation framework should include disease-scope refinement, literature review, mechanistic assessment, compound availability checks, and assay design.

1. Refine the disease context from pan-cancer to one or more specific tumor types, molecular subtypes, or mutation-defined contexts.
2. Review high-ranking candidates against approved indications, safety profiles, known mechanisms, literature support, novelty, and experimental feasibility.
3. Select a short list of candidate drug-target pairs for experimental assay design, prioritizing candidates with convergent evidence across interaction prediction, disease relevance, and structural support.
4. Use docking poses and target biology to guide experimental readouts, including target engagement, pathway response, cell viability, and dose-response assays.

5. Incorporate experimental results into the prioritization framework to improve future candidate selection.

11. Conclusion

The current BioMaster workflow has completed a report-scale computational screen of 915 approved small molecules against 1000 human protein targets, covering 915000 drug-target pairs. The workflow integrates AI-DTI prediction, target-disease evidence, protein network evidence, drug-disease graph evidence, and structural docking enhancement into a ranked candidate-prioritization system.

The resulting candidate set provides a structured basis for expert review and downstream validation. Its main value is not to claim efficacy directly, but to organize multiple evidence layers into a practical decision framework for drug repurposing research.